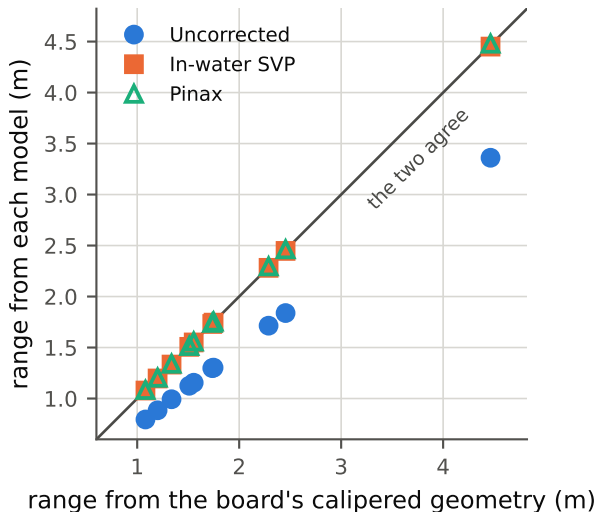


Two independent routes to range, and one that disagrees with both



Ten pool frames. Every range is absolute, solved from the board's calipered geometry — the object supplies the metric scale, the camera model only the angles. The reference is the mean of the two corrected models, which share no data and differ by 0.6 %: one is fitted to underwater frames and holds no refraction theory, the other comes from the in-air calibration and never sees one. An uncorrected port sits 26 % below both.